JungKyoo Shin

Curriculum Vitae
Seoul, South Korea | Email: neo293@naver.com
Google Scholar: https://scholar.google.com/citations?user=BmBtnK4AAAAJ

Education

Chung-Ang University | Ph.D. Candidate in **Artificial Intelligence**, 2022–Present
Dissertation: Bridging Multi-Level Video-Language Mismatch via Semantic and Temporal Representations
Advisor: Prof. Eunwoo Kim

University of Science and Technology / ETRI | M.S. in **Information Communication Technology**, 2019–2021
Thesis: Cross-modal Interaction Based Temporal Moment Localization in a Video
Advisor: Prof. Jinyoung Moon

Dongguk University | B.S. in **Information and Communication Engineering & Bioinformatics**, 2017–2019

Research Objectives

Keyword: Multi-modal understanding, Representation learning, Robot Learning

I am interested in how knowledge can be acquired, structured, and utilized from multimodal data. My research investigates **contextual understanding** across **images, videos, and natural language**, with a particular focus on representation learning for **semantic alignment, temporal reasoning, and knowledge transfer.** Through this perspective, I have studied diverse tasks including video retrieval, captioning, question answering, temporal moment localization, robot task planning and more. My long-term goal is to develop learning frameworks that are not restricted to specific tasks but instead enable AI systems to understand complex real-world observations, reason over acquired knowledge, and apply that knowledge to practical decision-making and action.

Publications

- **JungKyoo Shin**, Bumsoo Kim, Eunwoo Kim. “Generative Modeling of Class Probability for Multi-Modal Representation Learning.” **CVPR 2025, Highlight.**
- Kiseong Hong, **JungKyoo Shin**, Eunwoo Kim. “Stake the Points: Structure-Faithful Instance Unlearning.” **CVPR 2026.**
- **Jungkyoo Shin**, Sungmin Kang, Yoonsik Cho, Eunwoo Kim. “Dynamic Scale Position Embedding for Cross-Modal Representation Learning.” **Neural Networks, 2026.**
- **Jungkyoo Shin**, Jieun Han, SeungJun Kim, Yoonseon Oh, Eunwoo Kim. “Task Planning for Long-Horizon Cooking Tasks Based on Large Language Models.” **IROS 2024, Selected Oral.**
- Heunseung Lim, **Jungkyoo Shin**, Hyoungki Choi, Dohoon Kim, Eunwoo Kim, Joonki Paik. “Gravitated Latent Space Loss Generated by Metric Tensor for High-Dynamic Range Imaging.” **ICASSP 2024.**
- **Jungkyoo Shin**, Jinyoung Moon. “Learning to Combine the Modalities of Language and Video for Temporal Moment Localization.” **Computer Vision and Image Understanding, 2022.**
- **Jungkyoo Shin**, Jinyoung Moon. “Multi-Perspective Attention Network for Fast Temporal Moment Localization.” IEEE Access, 2021.
- **Jungkyoo Shin**, Jinyoung Moon. “Fast Temporal Information Retrieval in Videos with Visual Memory.” ICAIIC 2021.

- Hyunjoon Koo, **Jungkyoo Shin**, Eunwoo Kim. “Dual-Branch Scale Disentanglement for Text–Video Retrieval.” Pattern Recognition Letters, 2025.

International Patent

- **Method and System for Retrieving Video Segment by a Semantic Query.** US Patent No. 12,019,678, granted Jun. 25, 2024.
  Inventors: Jin Young Moon, **Jung Kyoo Shin**; Assignee: ETRI; Filed Aug. 4, 2022.

Projects

- **Samsung Research Funding & Incubation Center |** Learning Transferable Task Knowledge for Social Robots**, 2022-2024**
- **Hyundai Engineering & Construction Project |** AI-Based Wheel Loader Work-Mode Classification and Deformation Estimation Algorithm Development**, 2023**
- **ETRI DeepView |** High-Performance Visual Discovery Platform**, 2019-2021**

Teaching Experiences

Chung-Ang University | Capstone Design, 2022 Fall
Chung-Ang University | Machine Learning, 2023 Spring
Chung-Ang University | Advanced AI, 2024 Fall

Academic Services

Conference reviewer for CVPR, NeurIPS and ECCV.

Invited Talks

- The 1st Physical AI Research Group Workshop, 2026
- KRoC Flagship Conference, 2025
- AI Graduate School Symposium, 2025
- Chung-Ang University AI Symposium / AI Graduate School Academic Seminar, 2024